COMP 3721 **Assignment 3**

Note:

- Submit a single PDF file.
- Answer the questions according to textbook, lecture notes and class discussions, not ChatGPT or Google.

Q1:

For a target IP address of 14.12.72.8/24, find the following.

(a) Target IP address in binary

00001110.00001100.01001000.00001000/24

(b) Subnet mask in binary

11111111.11111111.11111111.00000000

(c) Subnet mask in dotted decimal

255.255.255.0

(d) First address (i.e., network address) of the block

14.12.72.0/24

(e) Last address (i.e., broadcast address) of the block

14.12.72.255/24

(f) First host address of the block

14.12.72.0 + 1 =

14.12.72.1/24

(g) Last host address of the block

14.12.72.255 - 1 =

14.12.72.254/24

(h) Number of addresses in the block

32- 24 = 8, 2^8= 256

(i) Maximum number of hosts in the block

Maximum hosts = Total addresses - 2 = 256 - 2 = 254

Q2:

For a target IP address of 200.107.16.17/18, find the following.

(a) Target IP address in binary

11001000.01101011.00010000.00010001/18

(b) Subnet mask in binary

11111111.11111111.11000000.00000000

(c) Subnet mask in dotted decimal

255.255.192.0

(d) First address (i.e., network address) of the block

200.107.0.0/18

(e) Last address (i.e., broadcast address) of the block

200.107.63.255/18

(f) First host address of the block

200.107.0.0 + 1 =

200.107.0.1/18

(g) Last host address of the block

200.107.63.255 - 1 =

200.107.63.254/18

(h) Number of addresses in the block

32-18= 14, 2^ 14 = 16384

(i) Maximum number of hosts in the block

Maximum hosts = Total addresses - 2 = 16384- 2 = 16382

Q3:

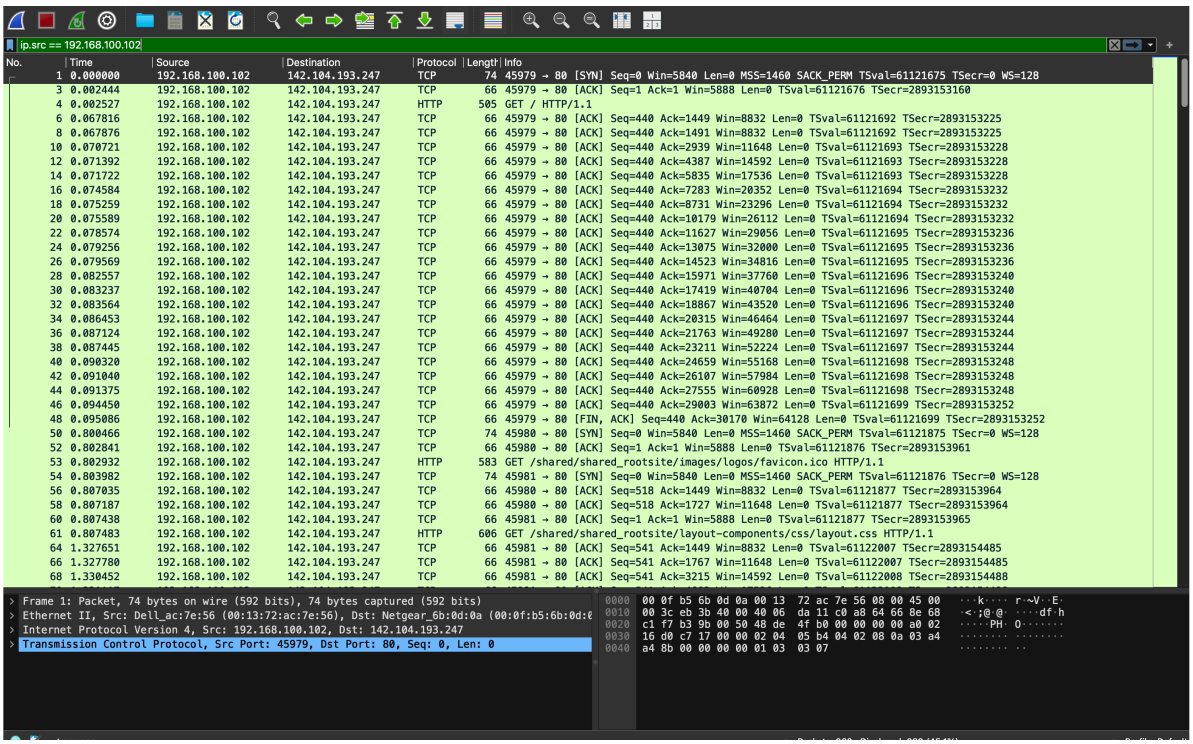
Download and install Wireshark (https://www.wireshark.org/). Open “trace.cap” (provided with the assignment) by either

- Double-click the capture file (if the extension is associated with Wireshark), or
- Open Wireshark, click File → Open → browse and choose the file.

Capture the Wireshark screen that shows the following by applying correct filters.

- One capture per question.
- Must show the display filter.
- Only need show the first few rows of the results, if any.

(a) Display all the packets with source address 192.168.100.102.



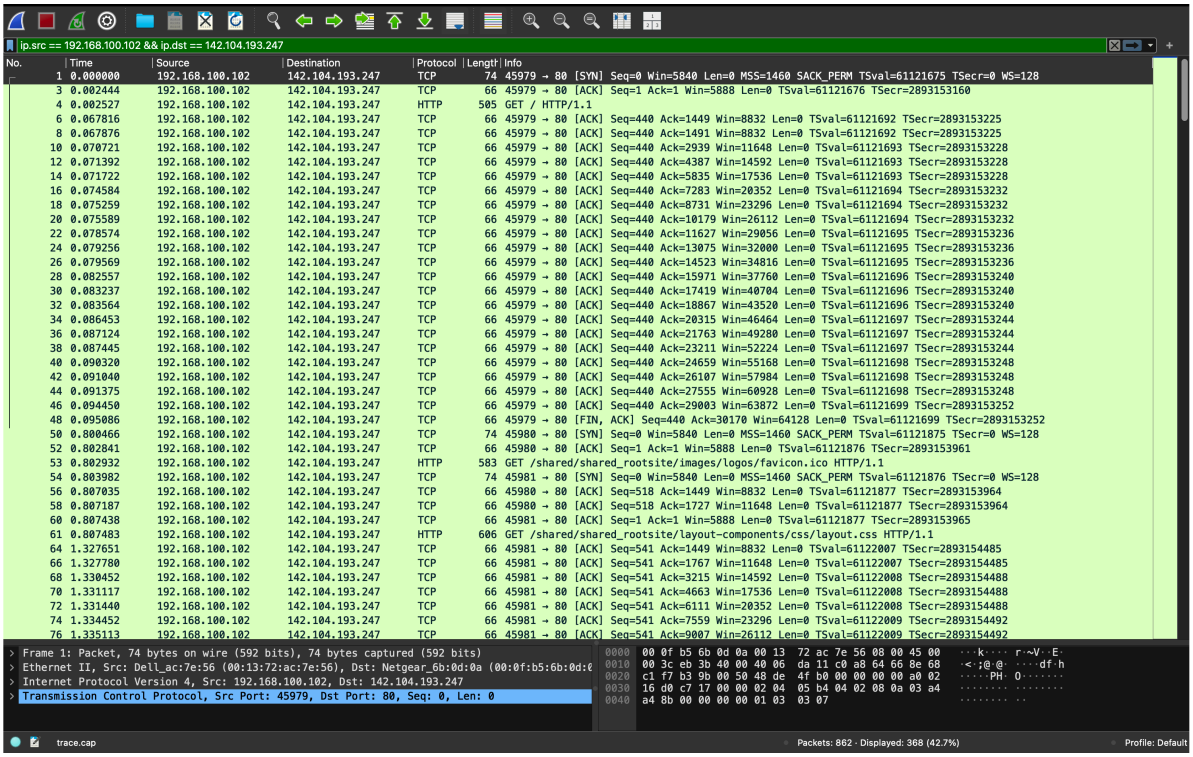
ip.src == 192.168.100.102

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.100.102	142.104.193.247	TCP	74	45979 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=61121675 TSecr=0 WS=128
3	0.002444	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=1 Ack=1 Win=5888 Len=0 TSval=61121676 TSecr=2893153160
4	0.002527	192.168.100.102	142.104.193.247	HTTP	505	GET / HTTP/1.1
6	0.067816	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=1449 Win=8832 Len=0 TSval=61121692 TSecr=2893153225
8	0.067876	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=1491 Win=8832 Len=0 TSval=61121692 TSecr=2893153225
10	0.070721	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=2939 Win=11648 Len=0 TSval=61121693 TSecr=2893153228
12	0.071392	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=4387 Win=14592 Len=0 TSval=61121693 TSecr=2893153228
14	0.071722	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=5835 Win=17536 Len=0 TSval=61121693 TSecr=2893153228
16	0.074584	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=7283 Win=20352 Len=0 TSval=61121694 TSecr=2893153232
18	0.075259	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=8731 Win=23296 Len=0 TSval=61121694 TSecr=2893153232
20	0.075589	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=10179 Win=26112 Len=0 TSval=61121694 TSecr=2893153232
22	0.078574	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=11627 Win=29056 Len=0 TSval=61121695 TSecr=2893153236
24	0.079256	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=13075 Win=32000 Len=0 TSval=61121695 TSecr=2893153236
26	0.079569	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=14523 Win=34816 Len=0 TSval=61121695 TSecr=2893153236
28	0.082557	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=15971 Win=37760 Len=0 TSval=61121696 TSecr=2893153240
30	0.083237	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=17419 Win=40704 Len=0 TSval=61121696 TSecr=2893153240
32	0.083564	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=18867 Win=43520 Len=0 TSval=61121696 TSecr=2893153240
34	0.086453	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=20315 Win=46464 Len=0 TSval=61121697 TSecr=2893153244
36	0.087124	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=21763 Win=49280 Len=0 TSval=61121697 TSecr=2893153244
38	0.087445	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=23211 Win=52224 Len=0 TSval=61121697 TSecr=2893153244
40	0.090320	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=24659 Win=55168 Len=0 TSval=61121698 TSecr=2893153248
42	0.091040	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=26107 Win=57984 Len=0 TSval=61121698 TSecr=2893153248
44	0.091375	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=27555 Win=60928 Len=0 TSval=61121698 TSecr=2893153248
46	0.094450	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [ACK] Seq=440 Ack=29003 Win=63872 Len=0 TSval=61121699 TSecr=2893153252
48	0.095086	192.168.100.102	142.104.193.247	TCP	66	45979 → 80 [FIN, ACK] Seq=440 Ack=30170 Win=64128 Len=0 TSval=61121699 TSecr=2893153252
50	0.800466	192.168.100.102	142.104.193.247	TCP	74	45980 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=61121875 TSecr=0 WS=128
52	0.802841	192.168.100.102	142.104.193.247	TCP	66	45980 → 80 [ACK] Seq=1 Ack=1 Win=5888 Len=0 TSval=61121876 TSecr=2893153961
53	0.802932	192.168.100.102	142.104.193.247	HTTP	583	GET /shared/shared_rootsite/images/logos/favicon.ico HTTP/1.1
54	0.803982	192.168.100.102	142.104.193.247	TCP	74	45981 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=61121876 TSecr=0 WS=128
56	0.807035	192.168.100.102	142.104.193.247	TCP	66	45980 → 80 [ACK] Seq=518 Ack=1449 Win=8832 Len=0 TSval=61121877 TSecr=2893153964
58	0.807187	192.168.100.102	142.104.193.247	TCP	66	45980 → 80 [ACK] Seq=518 Ack=1727 Win=11648 Len=0 TSval=61121877 TSecr=2893153964
60	0.807438	192.168.100.102	142.104.193.247	TCP	66	45981 → 80 [ACK] Seq=1 Ack=1 Win=5888 Len=0 TSval=61121877 TSecr=2893153965
61	0.807483	192.168.100.102	142.104.193.247	HTTP	606	GET /shared/shared_rootsite/layout-components/css/layout.css HTTP/1.1
64	1.327651	192.168.100.102	142.104.193.247	TCP	66	45981 → 80 [ACK] Seq=541 Ack=1449 Win=8832 Len=0 TSval=61122007 TSecr=2893154485
66	1.327780	192.168.100.102	142.104.193.247	TCP	66	45981 → 80 [ACK] Seq=541 Ack=1767 Win=11648 Len=0 TSval=61122007 TSecr=2893154485
68	1.330452	192.168.100.102	142.104.193.247	TCP	66	45981 → 80 [ACK] Seq=541 Ack=3215 Win=14592 Len=0 TSval=61122008 TSecr=2893154488

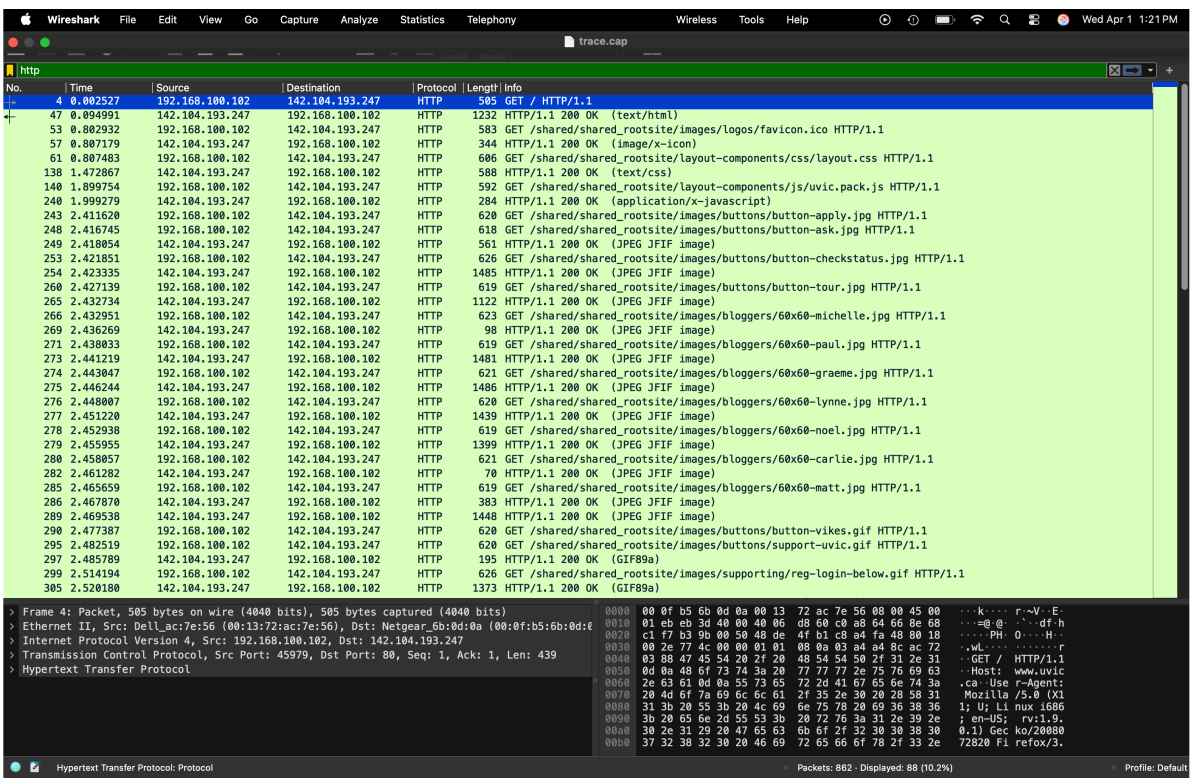
```
> Frame 1: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits)
> Ethernet II, Src: Dell_ac:7e:56 (00:13:72:ac:7e:56), Dst: Netgear_6b:0d:0a (00:0f:b5:6b:0d:0
> Internet Protocol Version 4, Src: 192.168.100.102, Dst: 142.104.193.247
> Transmission Control Protocol, Src Port: 45979, Dst Port: 80, Seq: 0, Len: 0

0000  00 0f b5 6b 0d 0a 00 13  72 ac 7e 56 08 00 45 00   ···k···· r·~V··E·
0010  00 3c eb 3b 40 00 40 06  da 11 c0 a8 64 66 8e 68   ·<·;@·@· ····df·h
0020  c1 f7 b3 9b 00 50 48 de  4f b0 00 00 00 00 a0 02   ····· PH· O·····
0030  16 d0 c7 17 00 00 02 04  05 b4 04 02 08 0a 03 a4   ····· ····· ····
0040  a4 8b 00 00 00 00 01 03  03 07                     ····· ··
```

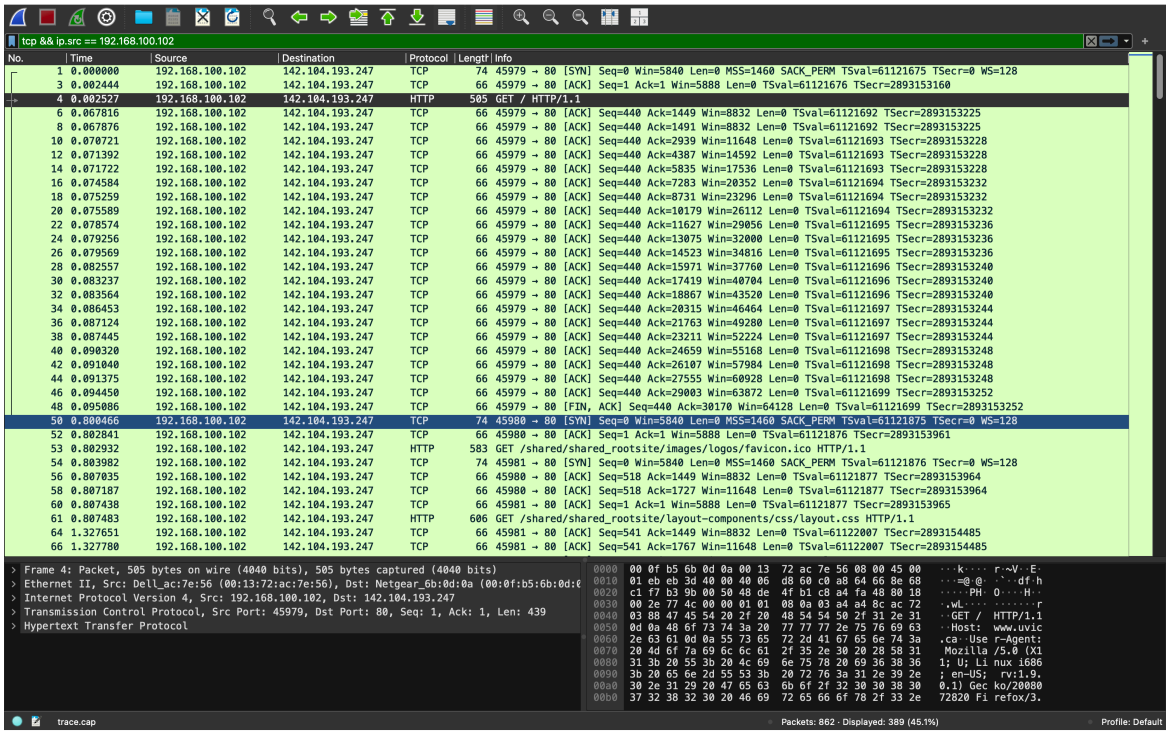
(b) Display all the packets with source address: 192.168.100.102 and destination address: 142.104.193.247.



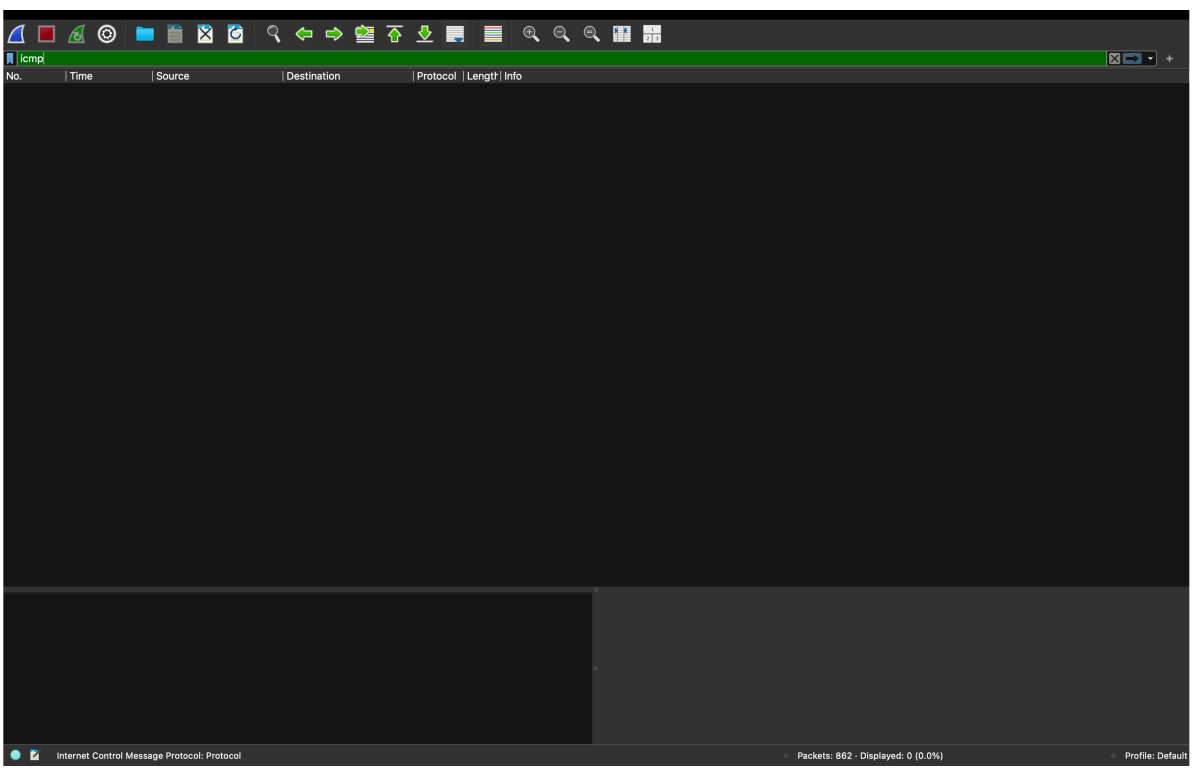
(c) Display all the HTTP packets.



(d) Display all the TCP packets with source address 192.168.100.102.



(e) Display all ICMP packets.



Appendix – Some Wireshark Display Filters:

- Comparing rules
  - English: `eq`, `le`, `ge`, `lt`, `gt`, `ne`
  - C-like: ==, <=, >=, <, >, `!`
- Logical operators
  - English: `and`, `or`, `xor`, `not`
  - C-like: `&&`, `||`, `^^`, `!`
- Boolean – when the field is present, the value is true.
  - For example, `tcp.flags.syn` is present in all TCP packets containing the flag, whether the SYN flag is 0 or 1. To only match TCP packets with the SYN flag set, you need to use `tcp.flags.syn == 1` or `tcp.flags.syn == True`.
- Addresses
  - IPv4 address: e.g., `ip.addr == 192.168.0.1`
  - IPv4 source address: e.g., display packets from 192.168.100.102 → `ip.src == 192.168.100.102`
  - Ethernet (link layer/MAC) address. e.g., `eth.dst == ff:ff:ff:ff:ff:ff`

For more information, consult Wireshark documentations on display filter expressions (https://www.wireshark.org/docs/wsug_html_chunked/ChWorkBuildDisplayFilterSection.html).

— End of Assignment 3 —